Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/26/2026
PI Name:	Haus, Miranda Ph.D.		
Project Title:	Investigating Legume Crop Rotations and Microbial Interactions in Regolith Simulants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 Space Biology NNH24ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	04/01/2026	**End Date:**	03/31/2030
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Horgan, Briony Ph.D. (Purdue University) Thompson, Michelle Ph.D. (Purdue University) Wilhelm, Roland Ph.D. (Purdue University)		
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Task Description:	The long-term goal of the proposed research is to advance our fundamental understanding of how legumes and plant-associated bacteria can thrive in extraterrestrial environments – knowledge that will pave the way for practical, space-based agriculture in future missions. Legumes build soil nitrogen (N) which is integral to conventional crop rotations. This proposal aims to: (1) characterize how plants and their associated microbes cope with the harsh conditions of regolith across a full crop rotation, and (2) identify methods to promote phytoremediation of regolith for soil, including mineral weathering and metal uptake. By facilitating and monitoring improvements to regolith soil fertility, and aligning the right plant with the right stage of regolith development or microbial partner, this research will help develop effective cropping systems for off-planet agriculture. Prior research compared the growth and physiology of alfalfa (Medicago sativa) grown in lunar regolith simulants: JSC-1A and OPRH4W30. Alfalfa grown in OPRH4W30 experienced greater stress and lower survival rates than JSC-1A. Selected endophytic microbes mitigated stress and recovered survival rates, such as a strain of non-nodulating, salinity-tolerant Paraburkholderia, but not a strain of the root-nodulating heavy metal tolerant Ensifer. Recovery was associated with higher root exudation and altered root architecture as evidenced by thinner roots, root branching reduction, and altered orientation of roots. These results suggest that plant-microbe interactions in regolith may not conform to the expected mutualistic dynamics observed in terrestrial soils. Building on this data, this project has three main objectives. 1) Examine how microbial partners regulate root physiology and architecture. Specifically, we will determine how microbial partners caused higher exudate leakage and whether more exudation accelerated the chemical weathering of the regolith simulant. Microbes that degrade lignin, metabolize ethylene precursors, or alter root hormone signaling may further influence how roots orient themselves in hostile substrates. 2) Test the ability to complete rotations of M. truncatula and wheat (T. aestivum) in simulant, studying the cyclical conditioning of soils (minerals and organo-mineral formation) and plants (nutrient, metal, and microbiome). Transcriptomics analyses of root tissues will reveal the key stressors and regulators of rooting exudation and growth, as well as endophyte gene expression. 3) Test additional regolith simulants, such as those containing abrasive glass shards or agglutinates, to determine whether the same strategies are generalizable to other planetary surfaces, including Martian regolith simulants. This work will advance our understanding of how crop rotations, and microbial partners, can foster soil development and plant performance in extreme soil conditions, informing the development of off-planet, regolith-based cropping systems. By combining molecular tools, advanced imaging, and soil chemistry analyses, our team will explore plant-microbe-regolith dynamics in settings where conventional farming is impossible. Outcomes will include a basic understanding of how root-microbe-soil dynamics can enhance regolith weathering, as well as an applied understanding of the feasibility of building soil fertility through crop rotations. These goals align with NASA's Space Crops Research goal and Bioastronautics and Life Support Systems (BLiSS) campaign, addressing multiple space-relevant stressors and accelerating our progress toward a bioregenerative life support system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research pioneers scientific discovery by revealing how plant-microbe partnerships drive regolith remediation and root adaptation under space-relevant stressors, enabling the development of self-sustaining cropping systems for off world missions. Expected outcomes include mechanistic insight into how crop rotations influence substrate fertility, identification of microbial traits that promote plant resilience, and understanding of how regolith properties shape biological performance across generations. Deliverables will include a comprehensive dataset linking plant, microbial, and mineral responses; optimized legume-microbe pairings for regolith agriculture; and validated imaging and molecular protocols for monitoring plant health and regolith transformation in space-based systems.
Task Progress:	New Project for FY2026
Bibliography Type:	Description: (Last Updated: 07/31/2025)